



Server Implementation of E-Agriculture

Group 6

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This paper provides a detailed overview of the configuration of web servers, database servers, DNS servers, and DHCP server applications on the Red Hat Enterprise Linux 9 platform. The authors have carefully examined various aspects of web server technology, DHCP servers, DNS servers, and database servers among others.

Hopefully this project can provide benefits and become a reference for further development in the field of network and system security.

Depok, 1 March 2025

Authors

SYSTEM ANALYSIS

This paper, "Server Implementation of E-Agriculture," discusses the integration of web servers, DHCP servers, database servers, and DNS servers in technology-based agricultural systems. This research aims to investigate and configure these five servers using Red Hat Enterprise Linux (RHEL). With proper integration, these servers can operate efficiently to enhance performance, communication between server, and distribute IP addresses for IoT devices in the E-Agriculture sector.

This server integration facilitates the development of an E-Agriculture network that Supporting the utilization of Internet of Things (IoT) devices. However, the main issue in this implementation is ensuring that all four servers can communicate effectively without any obstacles. The DHCP server must be able to dynamically distribute IP addresses for IoT devices. The DNS Server plays a role in translating domain names into IP addresses; here, the author implements Forward and Reverse Zones, facilitating more efficient communication between the servers and IoT clients. The Database Server is responsible for storing user or agricultural user data. The web server provides web pages for e-agriculture that can be accessed by various users. In addition, the system must implement secure firewall security rules to restrict access only to necessary services and ensure that each server operates according to its role, with firewall configurations defined. This is the service provided for each server:

1. DNS can be access, cockpit, dhcpv6-client, dns, and ssh.
2. DHCP can be access, cockpit, dhcpv6-client, dhcp, and ssh.
3. WEB can be access, cockpit, dhcpv6-client, http, https, and ssh.
4. Database can be access, cockpit, dhcpv6-client, and ssh.

The implementation process of this system includes several important stages, starting from the installation and configuration of each server to testing connectivity, security, and performance. Connectivity tests are conducted to ensure that all servers can communicate effectively, while security tests are performed by examining firewall configurations to prevent unauthorized access.

Configuration (DHCP Server)

1. Installing DHCP Server.

Before we configure DHCP server first we have to install DHCP server using dnf with “**dnf install dhcp-server -y**”.

```
Red Hat Enterprise Linux 9 for x86_64 - AppStream 9.4 kB/s | 4.5 kB 00:00
Red Hat Enterprise Linux 9 for x86_64 - AppStream 3.2 MB/s | 52 MB 00:16
Red Hat Enterprise Linux 9 for x86_64 - BaseOS 6.5 kB/s | 4.1 kB 00:00
Red Hat Enterprise Linux 9 for x86_64 - BaseOS 3.2 MB/s | 49 MB 00:15
Last metadata expiration check: 0:00:05 ago on Sun 23 Mar 2025 09:48:49 AM EDT.
Dependencies resolved.
=====
Package Arch Version Repository Size
=====
Installing:
dhcp-server x86_64 12:4.4.2-19.b1.el9 rhel-9-for-x86_64-baseos-rpms 1.2 M
Installing dependencies:
dhcp-common noarch 12:4.4.2-19.b1.el9 rhel-9-for-x86_64-baseos-rpms 131 k
Transaction Summary
=====
Install 2 Packages
Total download size: 1.4 M
Installed size: 4.2 M
Downloading Packages:
(1/2): dhcp-common-4.4.2-19.b1.el9.noarch.rpm 214 kB/s | 131 kB 00:00
(2/2): dhcp-server-4.4.2-19.b1.el9.x86_64.rpm 1.8 MB/s | 1.2 MB 00:00
-----
Total 2.0 MB/s | 1.4 MB 00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
Preparing : dhcp-common-12:4.4.2-19.b1.el9.noarch 1/1
Installing : dhcp-common-12:4.4.2-19.b1.el9.noarch 1/2
Running scriptlet: dhcp-server-12:4.4.2-19.b1.el9.x86_64 2/2
Installing : dhcp-server-12:4.4.2-19.b1.el9.x86_64 2/2
Running scriptlet: dhcp-server-12:4.4.2-19.b1.el9.x86_64 2/2
Verifying : dhcp-common-12:4.4.2-19.b1.el9.noarch 1/2
Verifying : dhcp-server-12:4.4.2-19.b1.el9.x86_64 2/2
Installed products updated.
Installed:
dhcp-common-12:4.4.2-19.b1.el9.noarch dhcp-server-12:4.4.2-19.b1.el9.x86_64
Complete!
```

2. Set a Static IP for the DHCP Server.

After do installation process we set Static IP for DHCP Server using command:
nmcli connection modify enp1s0 ipv4.addresses 192.168.10.3/24 ipv4.method manual
nmcli connection up enp1s0

After that ensure the IP for the DHCP Server has been verified using the command “**ip a show enp1s0**”.

```
[admin@localhost ~]$ ip a show enp1s0
2: enp1s0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 52:54:00:10:aa:bb brd ff:ff:ff:ff:ff:ff
    inet 192.168.10.3/24 brd 192.168.10.255 scope global noprefixroute enp1s0
        valid_lft forever preferred_lft forever
    inet6 fe80::5054:ff:fe10:aabb/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

Configuration (DHCP Server)

3. Configuring the DHCP Server.

Use **nano** to edit the **dhcpd.conf** file so that the administrator can configure the DHCP Server services as needed with commands “**sudo nano /etc/dhcp/dhcpd.conf**”.

Inside the **dhcpd.conf**, enter the IP address to be managed and enter the range.

```
subnet 192.168.10.0 netmask 255.255.255.0 {  
    range 192.168.10.100 192.168.10.200;  
}
```

Then we enable, and start DHCP service.

```
systemctl enable dhcp  
systemctl start dhcp
```

checking dhcp server status with “**systemctl status dhcp**” to ensure dhcp server is running.

```
[admin@localhost ~]$ systemctl status dhcpd  
● dhcpd.service - DHCPv4 Server Daemon  
   Loaded: loaded (/usr/lib/systemd/system/dhcpd.service; enabled; preset: disabled)  
   Active: active (running) since Sun 2025-03-23 02:39:07 WIB; 2min 59s ago  
     Docs: man:dhcpd(8)  
           man:dhcpd.conf(5)  
  Main PID: 982 (dhcpd)  
    Status: "Dispatching packets..."  
   Tasks: 1 (limit: 10948)  
  Memory: 6.4M  
     CPU: 8ms  
   CGroup: /system.slice/dhcpd.service  
           └─982 /usr/sbin/dhcpd -f -cf /etc/dhcp/dhcpd.conf -user dhcpd -group dhcpd --no-pid
```

After start DHCP service allows DHCP traffic in the firewall “**firewall-cmd --add-service=dhcp --permanent**” and continued with “**firewall-cmd --reload**”, so that the client will get the ip from the DHCP server.

Configuration (DHCP Server)

4. Test DHCP from Client.

Trying to conduct a test from the client to the DHCP using these command:

```
nmcli con modify enp1s0 ipv4.method auto
```

```
nmcli con down enp1s0 && nmcli con up enp1s0
```

after executing these command, then continue with the command “**ip a show enp1s0**” to check if an IP address has been received from the DHCP Server. then it will produce output from IoT client:

IoT1 (192.168.10.100)

```
[admin@localhost ~]$ ip a show enp1s0
2: enp1s0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 52:54:00:b6:5d:d6 brd ff:ff:ff:ff:ff:ff
    inet 192.168.10.100/24 brd 192.168.10.255 scope global dynamic noprefixroute enp1s0
        valid_lft 34133sec preferred_lft 34133sec
    inet6 fe80::5054:ff:feb6:5dd6/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

IoT2 (192.168.10.101)

```
[admin@localhost ~]$ ip a show enp1s0
2: enp1s0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 10
    link/ether 52:54:00:52:41:a7 brd ff:ff:ff:ff:ff:ff
    inet 192.168.10.101/24 brd 192.168.10.255 scope global dynamic noprefixroute enp1s0
        valid_lft 34173sec preferred_lft 34173sec
    inet6 fe80::5054:ff:fe52:41a7/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

Configuration (DB Server)

1. Installing httpd.

Before we configure Web server first we install httpd using dnf with command “**dnf install httpd -y**”.

```
[root@localhost ~]# dnf install httpd -y
Updating Subscription Management repositories.
Last metadata expiration check: 14:11:08 ago on Sun 23 Mar 2025 18:46:46 AM EDT.
Package httpd-2.4.62-1.el9_5.2.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[root@localhost ~]# systemctl status httpd
● httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; disabled; preset: disabled)
   Active: active (running) since Mon 2025-03-24 09:51:41 EDT; 26s ago
     Docs: man:httpd.service(8)
   Main PID: 3661 (httpd)
   Status: "Total requests: 0; Idle/Busy workers 100/0; Requests/sec: 0; Bytes served/sec: 0 B/sec"
    Tasks: 177 (limit: 22796)
   Memory: 40.2M
      CPU: 354ms
   CGroup: /system.slice/httpd.service
           └─3661 /usr/sbin/httpd -DFOREGROUND
             └─3662 /usr/sbin/httpd -DFOREGROUND
               └─3666 /usr/sbin/httpd -DFOREGROUND
                 └─3667 /usr/sbin/httpd -DFOREGROUND
                   └─3668 /usr/sbin/httpd -DFOREGROUND

Mar 24 09:51:41 localhost.localdomain systemd[1]: Starting The Apache HTTP Server...
Mar 24 09:51:41 localhost.localdomain httpd[3661]: AH00558: httpd: could not reliably determine the server's fully qualified domain name, using localhost.localdomain. Set the 'ServerName' directive globally to suppress this message
Mar 24 09:51:41 localhost.localdomain httpd[3661]: Server configured, listening on: port 80
Mar 24 09:51:41 localhost.localdomain systemd[1]: Started The Apache HTTP Server.
```

2. Allow HTTPD Traffic In the Firewall.

If the web server can be accessed from the network, we need to add firewall rules to allow HTTPD traffic, by using command:

```
firewall-cmd -permanent -add-service=http  
firewall-cmd -reload
```

3. Assign Static IP for WEB Server.

To ensure that the server has a fixed IP address, we can assign a static IP address using nmcli. This is done with the command:

```
nmcli connection modify enp1s0 ipv4.addresses 192.168.10.4/24 ipv4.method manual  
nmcli connection up enp1s0
```

4. Start and Enable HTTPD service.

After doing all process then activating the HTTPD service with “**systemctl enable httpd**” and start the service “**systemctl start httpd**”.

```
● httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
   Active: active (running) since Sun 2025-03-23 02:49:24 WIB; 10min ago
     Docs: man:httpd.service(8)
   Main PID: 920 (httpd)
   Status: "Total requests: 5; Idle/Busy workers 100/0; Requests/sec: 0.00808; Bytes served/sec: 31 B/sec"
    Tasks: 178 (limit: 10948)
   Memory: 11.9M
      CPU: 536ms
   CGroup: /system.slice/httpd.service
           └─920 /usr/sbin/httpd -DFOREGROUND
             └─973 /usr/sbin/httpd -DFOREGROUND
               └─974 /usr/sbin/httpd -DFOREGROUND
                 └─977 /usr/sbin/httpd -DFOREGROUND
                   └─978 /usr/sbin/httpd -DFOREGROUND
                     └─981 /usr/sbin/httpd -DFOREGROUND
```

As stated in the image, the HTTPD service is already running.

Configuration (Web Server)

5. Editing HTML File and opening the web

Once again, it's important to note that the script is not original from the author, it's automatically from Red Hat Enterprise Linux. The reader can create the **HTML** page as they wish.

```
1<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.1//EN" "http://www.w3.org/TR/xhtml11/DTD/xhtml11.dtd">
2
3<html xmlns="http://www.w3.org/1999/xhtml" xml:lang="en">
4  <head>
5    <title>Test Page for the HTTP Server on Red Hat Enterprise Linux</title>
6    <meta http-equiv="Content-Type" content="text/html; charset=UTF-8" />
7    <style type="text/css">
8      /**/
9      body {
10        background-color: #fff;
11        color: #000;
12        font-size: 1.1em;
13        font-family: "Red Hat Text", Helvetica, Tahoma, sans-serif;
14        margin: 0;
15        padding: 0;
16        border-bottom: 30px solid black;
17        min-height: 100vh;
18        box-sizing: border-box;
19      }
20      :link {
21        color: #c00;
22      }
23      :visited {
24        color: #c00;
25      }
26      a:hover {
27        color: #f50;
28      }
29      h1 {
30        text-align: left;
31        margin: 0;
32        margin-bottom: .6em;
33        padding: 1em 2em 1.5em 2em;
34        background-color: black;
35        color: #fff;
36        font-weight: normal;
37        font-size: 2.5em;
38        border-bottom: 2px solid #000;
39      }
40    &lt;/style&gt;
41  &lt;/head&gt;
42  &lt;body&gt;
43    &lt;h1&gt;Red Hat Enterprise Linux Test Page&lt;/h1&gt;
44  &lt;/body&gt;
45&lt;/html&gt;</pre></div><div data-bbox="140 546 865 601" data-label="Text"><p>After editing the <b>HTML</b> file, please use a browser to open the web page and enter the IP address of the web server. After entering the IP in the search icon, the web page will appear according to the created <b>HTML</b> file.</p></div><div data-bbox="142 601 853 680" data-label="Image"><img alt="Screenshot of a web browser showing the Red Hat Enterprise Linux Test Page. The page has a black header with the Red Hat logo and the text 'Red Hat Enterprise Linux Test Page' in white. The main content area is white with a black border at the bottom."/></div><div data-bbox="172 694 723 704" data-label="Text"><p>This page is used to test the proper operation of the HTTP server after it has been installed. If you can read this page, it means that the HTTP server installed at this site is working properly.</p></div><div data-bbox="184 717 333 727" data-label="Section-Header"><h4>If you are a member of the general public:</h4></div><div data-bbox="184 728 443 743" data-label="Text"><p>The fact that you are seeing this page indicates that the website you just visited is either experiencing problems, or is undergoing routine maintenance.</p></div><div data-bbox="184 744 479 764" data-label="Text"><p>If you would like to let the administrators of this website know that you've seen this page instead of the page you expected, you should send them e-mail. In general, mail sent to the name "webmaster" and directed to the website's domain should reach the appropriate person.</p></div><div data-bbox="184 766 476 780" data-label="Text"><p>For example, if you experienced problems while visiting <a href="http://www.example.com">www.example.com</a>, you should send e-mail to "webmaster@example.com".</p></div><div data-bbox="184 780 441 795" data-label="Text"><p>For information on Red Hat Enterprise Linux, please visit the <a href="http://www.redhat.com">Red Hat, Inc. website</a>. The documentation for Red Hat Enterprise Linux is <a href="http://www.redhat.com">available on the Red Hat, Inc. website</a>.</p></div><div data-bbox="507 717 637 727" data-label="Section-Header"><h4>If you are the website administrator:</h4></div><div data-bbox="507 728 792 743" data-label="Text"><p>You may now add content to the webroot directory. Note that until you do so, people visiting your website will see this page, and not your content.</p></div><div data-bbox="507 744 803 769" data-label="Text"><p>For systems using the Apache HTTP Server: You may now add content to the directory <code>/usr/local/httpd/</code>. Note that until you do so, people visiting your website will see this page, and not your content. To prevent this page from ever being used, follow the instructions in the file <code>/etc/httpd/conf.d/welcome.conf</code>.</p></div><div data-bbox="507 770 803 785" data-label="Text"><p>For systems using NGINX: You should now put your content in a location of your choice and edit the root configuration directive in the <code>nginx</code> configuration file <code>/etc/nginx/nginx.conf</code>.</p></div><div data-bbox="508 794 612 817" data-label="Image"><img alt="Red Hat Enterprise Linux logo and a circular icon with a red arrow pointing to the right."/></div><div data-bbox="175 836 347 845" data-label="Text"><p><small>Red Hat is a registered trademark of Red Hat, Inc. in the United States and/or other countries.<br/>OpenSSL is a registered trademark of OpenSSL.org.</small></p></div>
```

Configuration (Database Server)

1. Installing and Running MariaDB Service.

Before configuring the database server, first installing MariaDB with dnf using the command “**dnf install mariadb-server -y**”.

```
[root@localhost 8061ce]# systemctl start mariadb
[root@localhost 8061ce]# systemctl enable mariadb
Created symlink /etc/systemd/system/mysql.service → /usr/lib/systemd/system/mariadb.service.
Created symlink /etc/systemd/system/mysqld.service → /usr/lib/systemd/system/mariadb.service.
Created symlink /etc/systemd/system/multi-user.target.wants/mariadb.service → /usr/lib/systemd/system/mariadb.service.
[root@localhost 8061ce]# dnf install mariadb-server -y
Updating Subscription Management repositories.
Last metadata expiration check: 0:05:05 ago on Mon 24 Mar 2025 07:18:19 AM EDT.
Package mariadb-server-3:10.5.27-1.el9_5.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
```

Next, installing MariaDB, activate and run the service on MariaDB using command “**systemctl start mariadb**” and continued with “**systemctl enable mariadb**”.

```
[admin@localhost ~]$ systemctl status mariadb
● mariadb.service - MariaDB 10.5 database server
   Loaded: loaded (/usr/lib/systemd/system/mariadb.service; enabled; preset: disabled)
   Active: active (running) since Sun 2025-03-23 03:16:58 WIB; 25s ago
     Docs: man:mariadb(8)
           https://mariadb.com/kb/en/library/systemd/
   Process: 901 ExecStartPre=/usr/libexec/mariadb-check-socket (code=exited, status=0/SUCCESS)
   Process: 962 ExecStartPre=/usr/libexec/mariadb-prepare-db-dir mariadb.service (code=exited, status=0/SUCCESS)
   Process: 1182 ExecStartPost=/usr/libexec/mariadb-check-upgrade (code=exited, status=0/SUCCESS)
  Main PID: 1026 (mariabdd)
    Status: "Taking your SQL requests now..."
     Tasks: 14 (limit: 10948)
    Memory: 87.0M
       CPU: 181ms
    CGroup: /system.slice/mariadb.service
            └─1026 /usr/libexec/mariabdd --basedir=/usr
```

2. Create a Database in MariaDB

Before we create the database, we enter into MariaDB using “**mysql -u root -p**” to manage the database. To log in to MariaDB using the **root** account with the command mentioned earlier.

```
[root@localhost 8061ce]# mysql -u root -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 3
Server version: 10.5.27-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> CREATE DATABASE farmdb;
Query OK, 1 row affected (0.002 sec)

MariaDB [(none)]> CREATE USER 'farmuser'@'%' IDENTIFIED BY 'StrongPassword123!'; GRANT ALL PRIVILEGES ON farmdb.* TO 'farmuser'@'%' ; FLUSH PRIVILEGES; EXIT;
Query OK, 0 rows affected (0.030 sec)

Query OK, 0 rows affected (0.002 sec)

Query OK, 0 rows affected (0.004 sec)

Bye
```

After login into MariaDB create database with “**CREATE DATABASE farmdb;**” then create a user for remote in MariaDB “**CREATE USER 'farmuser'@'%' IDENTIFIED BY 'StrongPassword123!'; GRANT ALL PRIVILEGES ON farmdb.* TO 'farmuser'@'%' ; FLUSH PRIVILEGES; EXIT;**”.

Configuration (Database Server)

3. Allow Remote Access.

Activate remote access so that the administrator can manage the database system by modifying changes to the `mariadb-server.cnf` configuration file with command “`gedit /etc/my.cnf.d/mariadb-server.cnf`”. Inside `mariadb-server.cnf` find the `bind-address` on the page line, then uncomment the `bind-address` by deleting “#” from “`#bind-address=0.0.0.0`”.

```
34 #
35 # Allow server to accept connections on all interfaces.
36 #
37 bind-address=0.0.0.0
38 #
39 # Optional setting
40 #wsrep_slave_threads=1
41 #innodb_flush_log_at_trx_commit=0
```

Next, please restart MariaDB due to configuration changes made in MariaDB using by “`systemctl restart mariadb`”.

4. Conducting a connection test to the MariaDB database from a virtual machine.

Next, perform a test to check the connection from another virtual machine to the previously created database, using the user `farmuser` from the previously created database “`farmdb`” to the other virtual machine. Use the command “`mysql -u farmuser -p -h db.local`”, then it will execute and produce the following output:

```
[admin@localhost ~]$ mysql -u farmuser -p -h db.local
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 3
Server version: 10.5.27-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> SHOW DATABASES;
+-----+
| Database |
+-----+
| farmdb   |
| information_schema |
+-----+
2 rows in set (0.003 sec)
```

Configuration (DNS Server)

1. Installing Bind

Before configuring DNS server we have to install the necessary packages first for the DNS server, use the command **"dnf install bind bind-utils -y"**.

```
Updating Subscription Management repositories.
Last metadata expiration check: 2:20:18 ago on Mon 24 Mar 2025 11:52:01 PM EDT.
Package bind-utils-32:9.16.23-24.el9_5.3.x86_64 is already installed.
Dependencies resolved.
=====
Package                Architecture Version                Repository              Size
=====
Installing:
bind                   x86_64          32:9.16.23-24.el9_5.3  rhel-9-for-x86_64-appstream-rpms 508 k
Installing dependencies:
bind-dnssec-doc        noarch          32:9.16.23-24.el9_5.3  rhel-9-for-x86_64-appstream-rpms 48 k
python3-bind           noarch          32:9.16.23-24.el9_5.3  rhel-9-for-x86_64-appstream-rpms 71 k
python3-ply            noarch          3.11-14.el9            rhel-9-for-x86_64-appstream-rpms 111 k
Installing weak dependencies:
bind-dnssec-utils      x86_64          32:9.16.23-24.el9_5.3  rhel-9-for-x86_64-appstream-rpms 122 k
Transaction Summary
=====
Install 5 Packages

Total download size: 860 k
Installed size: 2.5 M
Downloading Packages:
(1/5): bind-dnssec-doc-9.16.23-24.el9_5.3.noarch.rpm 59 kB/s | 48 kB 00:00
(2/5): python3-ply-3.11-14.el9.noarch.rpm 127 kB/s | 111 kB 00:00
(3/5): bind-9.16.23-24.el9_5.3.x86_64.rpm 525 kB/s | 508 kB 00:00
(4/5): bind-dnssec-utils-9.16.23-24.el9_5.3.x86_64.rpm 374 kB/s | 122 kB 00:00
(5/5): python3-bind-9.16.23-24.el9_5.3.noarch.rpm 242 kB/s | 71 kB 00:00
-----
Total 723 kB/s | 860 kB 00:01
```

After installing the necessary packages for the DNS server, the next step is to configure a static IP address on the DNS server. This is done using the command:

```
nmcli connection modify enp1s0 ipv4.addresses 192.168.10.2/24 ipv4.method manual
nmcli connection up enp1s0
```

2. Configuring named.conf File

Configure the **named.conf** file by adding features to ensure that the DNS service operates optimally and securely. The settings include listen-on port 53 to receive requests, allow-query to limit client access, recursion yes so that the server caches query results to speed up domain name lookups, and logging to record DNS activity. Here is the configure on **named.conf** file:

Configuration (DNS Server)

```
//
// named.conf
//
// Provided by Red Hat bind package to configure the ISC BIND named(8) DNS
// server as a caching only nameserver (as a localhost DNS resolver only).
//
// See /usr/share/doc/bind*/sample/ for example named configuration files.
//

options {
    listen-on port 53 { 127.0.0.1; 192.168.10.2; };
    listen-on-v6 port 53 { ::1; };
    directory "/var/named";
    dump-file "/var/named/data/cache_dump.db";
    statistics-file "/var/named/data/named_stats.txt";
    memstatistics-file "/var/named/data/named_mem_stats.txt";
    secroots-file "/var/named/data/named.secroots";
    recursing-file "/var/named/data/named.recursing";
    allow-query { localhost; 192.168.10.2; any; };
    allow-query-cache { localhost; 192.168.10.2; any; };

    /*
     - If you are building an AUTHORITATIVE DNS server, do NOT enable recursion.
     - If you are building a RECURSIVE (caching) DNS server, you need to enable
       recursion.
     - If your recursive DNS server has a public IP address, you MUST enable access
       control to limit queries to your legitimate users. Failing to do so will
       cause your server to become part of large scale DNS amplification
       attacks. Implementing BCP38 within your network would greatly
       reduce such attack surface
    */
    recursion yes;

    dnssec-validation yes;

    managed-keys-directory "/var/named/dynamic";
    geoip-directory "/usr/share/GeoIP";

    pid-file "/run/named/named.pid";
    session-keyfile "/run/named/session.key";

    /* https://fedoraproject.org/wiki/Changes/CryptoPolicy */
    include "/etc/crypto-policies/back-ends/bind.config";
};

logging {
    channel default_debug {
        file "data/named.run";
        severity dynamic;
    };
};

zone "dns.local" IN {
    type master;
    file "/etc/named/dns.local.rev";
};

zone "10.168.192.in-addr.arpa" IN {
    type master;
    file "/etc/named/dhcp.local.zone";
};

zone "web.local" {
    type master;
    file "/etc/named/web.local.zone";
};

zone "db.local" {
    type master;
    file "/etc/named/db.local.zone";
};

zone "." IN {
    type hint;
    file "named.ca";
};

include "/etc/named.rfc1912.zones";
include "/etc/named.root.key";
```

Configuration (DNS Server)

3. Forward and Reverse DNS Zone Configuration.

Once the configuration on the `named.conf` file is complete, the next step is to set the Forward Zone and Reverse Zone. Forward Zone is used to translate domain names into IP addresses, while Reverse Zones are the opposite. In this configuration, the Forward Zone will be applied to `dns.local`, `dhcp.local`, `db.local`, and `web.local`, while the Reverse Zone will be configured on `dns.local.rev`. Here are the configuration details:

- Forward Zone `dns.local`

```
$TTL 86400
@      IN      SOA      dns.local. root.dns.local. (
        2025032201      ;Serial
        3600            ;Refresh
        1800            ;Retry
        604800          ;Expire
        86400           ;Minimum TTL
)

@      IN      NS       ns1.dns.local.
@      IN      NS       ns2.dns.local.
@      IN      A        192.168.10.2
ns1     IN      A        192.168.10.2
ns2     IN      A        192.168.10.2
www     IN      CNAME    dns.local.
blog    IN      A        192.168.10.2
```

- Forward Zone `dhcp.local`

```
$TTL 86400
@      IN      SOA      dhcp.local. root.dhcp.local. (
        2025032201      ;Serial
        3600            ;Refresh
        1800            ;Retry
        604800          ;Expire
        86400           ;Minimum TTL
)

@      IN      NS       ns1.dhcp.local.
@      IN      NS       ns2.dhcp.local.
@      IN      A        192.168.10.3
ns1     IN      A        192.168.10.3
ns2     IN      A        192.168.10.3
www     IN      CNAME    dhcp.local.
blog    IN      A        192.168.10.3
```

Configuration (DNS Server)

- Forward Zone **db.local**

```
$TTL 86400
@      IN      SOA      db.local. root.db.local. (
        2025032201      ;Serial
        3600            ;Refresh
        1800            ;Retry
        604800          ;Expire
        86400           ;Minimum TTL
)

@      IN      NS       ns1.db.local.
@      IN      NS       ns2.db.local.
@      IN      A        192.168.10.5
ns1     IN      A        192.168.10.5
ns2     IN      A        192.168.10.5
www     IN      CNAME    db.local.
blog    IN      A        192.168.10.5
```

- Forward Zone **web.local**

```
$TTL 86400
@      IN      SOA      dhcp.local. root.web.local. (
        2025032201      ;Serial
        3600            ;Refresh
        1800            ;Retry
        604800          ;Expire
        86400           ;Minimum TTL
)

@      IN      NS       ns1.web.local.
@      IN      NS       ns2.web.local.
@      IN      A        192.168.10.4
ns1     IN      A        192.168.10.4
ns2     IN      A        192.168.10.4
www     IN      CNAME    web.local.
blog    IN      A        192.168.10.4
```

Configuration (DNS Server)

- Reverse Zone `dns.local.rev`

```
$TTL 86400
@      IN      SOA      dns.local. root.dns.local. (
        2025032201      ;Serial
        3600             ;Refresh
        1800             ;Retry
        604800           ;Expire
        86400            ;Minimum TTL
)

; Nameservers
@      IN      NS       ns1.dns.local.
@      IN      NS       ns2.dns.local.
; A records
@      IN      A        192.168.10.2
ns1    IN      A        192.168.10.2
ns2    IN      A        192.168.10.2
; PTR records
50     IN      PTR      dns.local.
50     IN      PTR      blog.dns.local.
51     IN      PTR      dhcp.local
52     IN      PTR      web.local
53     IN      PTR      db.local
```

Next, To ensure that the DNS service runs and activated automatically when the system is rebooted, use the command:

```
Systemctl enable named  
systemctl start named
```

To ensure that DNS connections are allowed, the necessary ports must be opened in the firewall. The following commands were executed:

```
firewall-cmd -add-service=dns -permanent  
firewall-cmd -add-service=named -permanent
```

4. Nameserver Configuration on Each Server

Next, for each server can communicate with one another using the configured DNS, it is necessary to set up the nameserver on each server. This is done by adding the IP address of the DNS server into the configuration file using “**gedit /etc/resolv.conf**”.



```
1 # Generated by NetworkManager
2 search dns.local
3 nameserver 192.168.10.2
```


Configuration (DNS Server)

5. Connection Testing

After, through all process we can testing the connection by entering the server domain. Before do connection testing, make sure that every server has the right ip routing so they can ping each other. since we're using 192.168.10.0 subnet, we're going to check the ip route using "**ip route**":

```
[root@localhost named]# ip route
192.168.10.0/24 dev enp1s0 proto kernel scope link src 192.168.10.2 metric 100
```

Next, after ensuring that each server has the correct IP routing using "**ip route**", attempt to ping using the domain address of the DHCP server ("**dhcp.local**") to the DNS Server ("**dns.local**"). There is network testing by its domain as shown in the image:

```
[admin@localhost ~]$ ping dns.local
PING dns.local (192.168.10.2) 56(84) bytes of data.
64 bytes from 192.168.10.2 (192.168.10.2): icmp_seq=1 ttl=64 time=0.352 ms
64 bytes from 192.168.10.2 (192.168.10.2): icmp_seq=2 ttl=64 time=0.474 ms
64 bytes from 192.168.10.2 (192.168.10.2): icmp_seq=3 ttl=64 time=0.536 ms
^C
--- dns.local ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2004ms
rtt min/avg/max/mdev = 0.352/0.454/0.536/0.076 ms
```

This is the list of services permitted through the firewall on each server within the network. This list includes services such as **DNS**, **DHCP**, **SSH**, **HTTP**, and **HTTPS**, which are necessary for server can be communicated. Each server has firewall rules that allows some services like:

- a. DNS:
 - cockpit
 - dhcpv6-client
 - dns
 - ssh
- b. DHCP:
 - cockpit
 - dhcpv6-client
 - dhcp
 - ssh
- c. WEB:
 - cockpit
 - dhcpv6-client
 - http
 - https
 - ssh
- d. Database
 - cockpit
 - dhcpv6-client
 - ssh

REQUIREMENTS

Hardware :

1. Lenovo Legion Slim 5I 16irh8

Operating System :

1. RedHat Enterprise Linux 9
2. Arch Linux

Software :

1. VMware Workstation Pro
2. Microsoft Word
3. Canva

PROJECT FILE DETAILS

No	File Name	Remarks
1.	LSC Project – Group 9 2CS2	PDF that contains report
2.	LSC Project – Group 9 2CS2	Canva as PDF that contains presentation slide
4.	LSC Project – Group 9 2CS2	Microsoft Word that contains report